

Igor Monteiro Magro

Data Engineering | Machine Learning | Kubernetes Infrastructure

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SUMMARY

Computer Engineering senior (IFF, Brazil) with 2 years of IT experience in the public sector, building production data pipelines and services with **Python, SQL, PostgreSQL, Docker, Kubernetes and Argo CD**. Hands-on with log ingestion and observability (OpenTelemetry, Grafana Loki, Grafana) and applied machine learning: in my thesis, anomaly detectors trained on 70 million network records caught 96% of the attacks the firewall missed. Advanced English, intermediate German. Seeking a Data Engineer, ML Engineer or Data Scientist role.

EXPERIENCE

IT Intern — Infrastructure and Data · City Hall of Campos dos Goytacazes, Brazil · Aug 2024 – Oct 2026

- Built a production data pipeline for DMARC reports (IMAP → XML parsing → PostgreSQL), running as a Kubernetes CronJob with GitOps delivery (GitLab CI, Kustomize, Argo CD, Sealed Secrets), replacing a paid service.
- Created a Metabase dashboard tracking ~85K emails over 4 days at 99.88% DMARC compliance; it revealed 0% DKIM alignment and guided the fix to the email policy.
- Developed an async API (FastAPI, SQLAlchemy, PostgreSQL) that centralizes MAC addresses blocked by MikroTik firewalls, with an auditable history and RouterOS sync scripts, replacing ticket-based unblocking with self-service through Rundeck.
- Deployed network log collection with OpenTelemetry, Promtail, Grafana Loki and Grafana for the MikroTik routers across municipal sites.
- Built an Active Directory event log pipeline (.evtx → Loki + PostgreSQL) for login/logoff audits, plus log collection for the Zimbra email server.
- Versioned and automated infrastructure service deployments with Docker, Kubernetes and Argo CD on a private cloud (CloudStack).

PROJECTS

Anomaly Detection in Network Logs (Undergraduate Thesis) · github.com/IgMnT/mikrotik-anomaly-detection

- Observability stack (OpenTelemetry, Loki, Grafana) collected 70 million firewall records in one week; a Docker lab generated labeled attacks as ground truth.
- Unsupervised detectors (LOF, DBSCAN, One-Class SVM, Isolation Forest) with Polars, scikit-learn and cuML (GPU): LOF caught 96% of 25 low-and-slow distributed attacks that the firewall rules missed.
- Near-real-time inference service raises alerts with a median latency of ~7 minutes.

Health Insurance Cross-Sell Prediction · github.com/IgMnT/Projeto01-PropensaoDeCompra

- Classification model that ranks customers by propensity to buy health insurance, served through a Flask REST API deployed on Render.

Food Delivery Business Dashboard · github.com/IgMnT/ProjetoFinalFTC

- Streamlit dashboard with the business metrics of a food delivery marketplace, deployed to the cloud and shared by public link.

SKILLS

- **Languages and data:** Python, SQL, pandas, Polars, PostgreSQL, MySQL, SQLite
- **Data engineering:** ETL/ELT, ingestion from APIs and IMAP, SQLAlchemy, Alembic, Metabase, Streamlit
- **Machine learning:** scikit-learn, cuML (GPU), MLflow, classification, regression, clustering, anomaly detection, cross-validation, hyperparameter tuning
- **Infrastructure and DevOps:** Docker, Docker Compose, Kubernetes, Kustomize, Argo CD, Sealed Secrets, GitLab CI, GitHub Actions, Linux
- **Observability:** OpenTelemetry, Grafana Loki, Promtail, Grafana · **APIs:** FastAPI, Flask, REST
- **Tools:** Git, GitHub, GitLab, AI coding assistants (Claude Code)

EDUCATION AND LANGUAGES

B.Sc. in Computer Engineering · Instituto Federal Fluminense (IFF), Brazil · expected 2026

Data Scientist Program · Comunidade DS · ~300 h · 2024–2025 — Machine Learning Fundamentals, SQL, Data Science in Production, Data Science on AWS and GCP, Business Metrics, and clustering, classification and regression projects

Languages: Portuguese (native) · English (advanced, professional working proficiency) · German (intermediate)